

Exact methods

Branch & Bound

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Optimization: principles and algorithms



ÉCOLE POLYTECHNIQUE
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Branch & Bound

In the absence of optimality conditions, enumeration is the only way to find the optimal solution.

However, it is most of the time impossible to perform explicitly due to the curse of dimensionality.

The branch & bound method is some sort of implicit enumeration technique, that attacks the huge set of feasible solutions using a “divide and conquer” strategy.

Combinatorial optimization

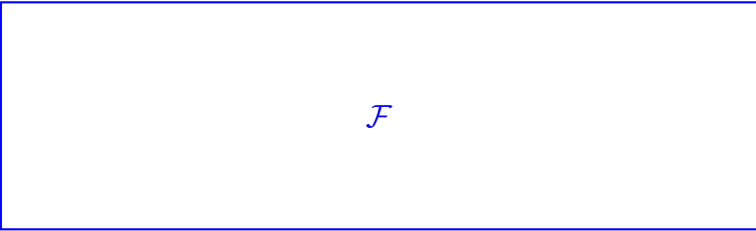
$$\min_x f(x)$$

subject to

$$x \in \mathcal{F},$$

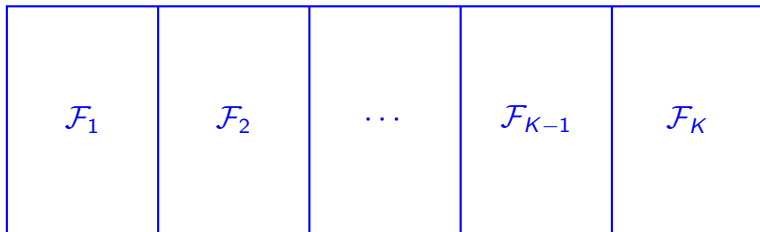
where $\mathcal{F}$ is a large set containing a finite number of elements.

Divide



$\mathcal{F}$

Divide



Divide

Problem P

$$\min_x f(x)$$

subject to

$$x \in \mathcal{F},$$

Solution: x^* .

Problem P_k

$$\min_x f(x)$$

subject to

$$x \in \mathcal{F}_k,$$

Solution: x_k^* .

Conquer

Theorem 26.1

Consider i such that

$$f(x_i^*) \leq f(x_k^*), \quad k = 1, \dots, K.$$

Then,

$$f(x^*) = f(x_i^*),$$

and x_i^* is solution of P .

$$f(x^*) \leq f(x_i^*)$$

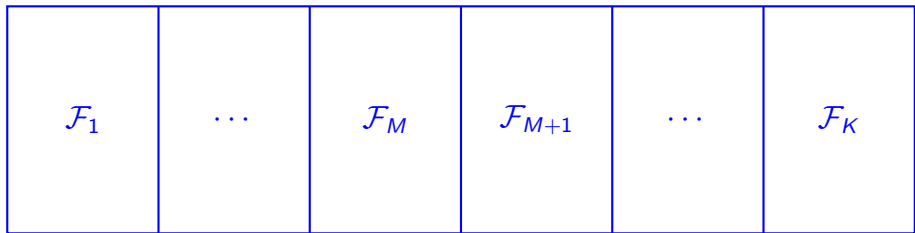
$\exists j$ such that $x^* \in \mathcal{F}_j$

Optimality of x_j^ :* $f(x_j^*) \leq f(x) \forall x \in \mathcal{F}_j$

$$x^* \in \mathcal{F}_j \Rightarrow f(x_j^*) \leq f(x^*)$$

$$f(x^*) \leq f(x_i^*) \leq f(x_j^*) \leq f(x^*)$$

Divide



Divide

Problem P

$$\min_x f(x)$$

subject to

$$x \in \mathcal{F}.$$

Solution: x^* .

Problem $P_k, k \leq M$

$$\min_x f(x)$$

subject to

$$x \in \mathcal{F}_k.$$

Solution: x_k^* .

Problem $P_k, k > M$

$$\min_x f(x)$$

subject to

$$x \in \mathcal{F}_k.$$

Lower bound:
 $\ell(P_m) \leq f(x_k^*).$

Conquer

Corollary 26.2

Consider i such that

$$f(x_i^*) \leq f(x_k^*), \quad k = 1, \dots, M,$$

and

$$f(x_i^*) \leq \ell(P_k), \quad k = M+1, \dots, K.$$

$$\ell(P_k) \leq f(x_k^*) \text{ for each } k > M$$

$$\text{Therefore, } f(x_i^*) \leq f(x_k^*) \text{ for all } k$$

The previous theorem applies.

Then,

$$f(x^*) = f(x_i^*),$$

and x_i^* is solution of P .

Summary

When the feasible set is large, one way to solve the problem is to partition it into smaller subsets

Solve each problem independently. The best solution found is the optimal solution of the original problem.

The most important result says that it is not necessary to solve all problems.

If good lower bounds can be found, it is sufficient to prove the optimality.

And for MILP, we know how to calculate lower bound: by solving the relaxation problem.